

Off-diagonal singularity of the current–current correlation measure: two explicit ergodic models

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Status. Candidate full counterexample to the broad formulation; likely valid subject to human review. A second, completely localized model shows that localization plus a Wegner estimate is still insufficient. Neither result settles the narrower i.i.d.-per-physical-site Anderson subclass.

1 Source question

Combes, Germinet, and Hislop [1] define the positive current–current correlation (ccc) measure by

$$M(\Delta_1, \Delta_2) = \mathcal{T}(vE_H(\Delta_1)vE_H(\Delta_2)), \quad v = i[H, X], \quad (1)$$

and ask on page 5 whether a density $m(E_1, E_2)$ exists when $E_1 \neq E_2$. They note the positive theorem of Bellissard–Hislop under strong disorder and an analytic single-site density [2].

measure $m(E, E)$ exists on the diagonal. If, in addition, $\ell(I_\epsilon(E)) \rightarrow 0$ as $\epsilon \rightarrow 0$, then we have

$$\lim_{\epsilon \rightarrow 0} \frac{M(I_\epsilon(E), I_\epsilon(E))}{\epsilon^2} = 0. \quad (17)$$

Consequently, $\sigma^{(DC)}(E) = 0$.

Finally, we mention the open question of the existence of a density $m(E_1, E_2)$ for $E_1 \neq E_2$ that we refer to as the off-diagonal case. This is important in view of the formula for the localization length (16), and in order to control the density in a neighborhood of the diagonal. Unfortunately, the approach developed in this note does not seem to allow us to control the off-diagonal behavior. We mention that for lattice models on $\ell^2(\mathbb{Z}^d)$, Bellissard and Hislop [6] proved the existence of a density $m_\alpha(E_1, E_2)$ at off-diagonal energies (E_1, E_2) provided that 1) the energies lie in a region away from the diagonal $E_1 = E_2$ determined by the strength of the disorder, and 2) the density of the single-site probability measure extends to a function analytic in a strip around the real axis.

2. Existence and Vanishing of the Density on the Diagonal in the Complete Localization Regime

In this section, we give a simple proof of Theorem 1 on the existence of, and vanishing of, the ccc-density on the diagonal $m(E, E)$ when $E \in \mathbb{R}^{CL}$. We

The unrestricted answer is negative for a scalar model with standard one-site covariance. A complementary unit-cell model retains complete localization, a bounded-density Wegner estimate, deterministic finite-range hopping, and a randomness-independent velocity.

2 Proof intuition

The first mechanism is Floquet-theoretic. A random choice between the two phases of a period-two potential is an ergodic family under ordinary lattice translations. Interband current transitions conserve quasi-momentum, so their two energy coordinates trace a curve, rather than filling the energy plane.

The second mechanism shows why localization does not repair the issue. Put one random scalar ω_n on a two-level cell. It shifts both cell energies together, while a fixed hopping keeps their gap equal to $2t$. The velocity exchanges the two eigenvectors, so averaging spreads ccc mass along two lines. In both cases a square of side ε captures order ε mass rather than order ε^2 mass.

3 An exact scalar counterexample with standard translations

Let $a, t, s > 0$ satisfy $2s > t^2/a$. On scalar $\ell^2(\mathbb{Z})$ define, for $q \in \{0, 1\}$,

$$(K_q \psi)(j) = s(\psi(j+2) + \psi(j-2)) + t(\psi(j+1) + \psi(j-1)) + a(-1)^{j+q} \psi(j). \quad (2)$$

Give the two values of q equal probability. Standard translation by one site sends K_q to $K_{q+1 \bmod 2}$, so this two-point family is covariant. The flip action is transitive and hence ergodic. Since only the diagonal potential changes with q , $i[K_q, X]$ is independent of q .

Theorem 1. *The ccc measure of (2) is not absolutely continuous on the off-diagonal. More strongly, at a suitable off-diagonal point (E_1^0, E_2^0) ,*

$$\frac{M([E_1^0, E_1^0 + \varepsilon], [E_2^0, E_2^0 + \varepsilon])}{\varepsilon^2} \longrightarrow +\infty.$$

Thus the unrestricted off-diagonal density assertion is false on scalar $\ell^2(\mathbb{Z})$ with the standard translation representation.

Proof. It is enough to use the phase $q = 0$, since the two phases are translates. In a two-site Floquet decomposition, with $k \in [-\pi, \pi]$,

$$K(k) = 2s \cos k I + d(k) \cdot \sigma, \quad d(k) = (t(1 + \cos k), t \sin k, a).$$

Put

$$R(k) = \sqrt{a^2 + 4t^2 \cos^2(k/2)}, \quad \lambda_{\pm}(k) = 2s \cos k \pm R(k).$$

In the same representation the velocity is

$$v(k) = -4s \sin k I + g(k) \cdot \sigma, \quad g(k) = (-t \sin k, -t(1 - \cos k), 0).$$

The scalar part has zero interband matrix element. For a two-level matrix, the squared interband matrix element of $g \cdot \sigma$ is $|g|^2 - (g \cdot d)^2/|d|^2$. Here $|g|^2 = 4t^2 \sin^2(k/2)$ and $g \cdot d = -2t^2 \sin k$, so direct simplification gives

$$w(k) := |\langle k, + | i[K, X] | k, - \rangle|^2 = \frac{4a^2 t^2 \sin^2(k/2)}{R(k)^2} > 0 \quad (0 < k < \pi). \quad (3)$$

Consequently the off-diagonal part of the ccc measure is, up to the harmless positive trace normalization,

$$\int_{-\pi}^{\pi} w(k) [\delta_{(\lambda_+(k), \lambda_-(k))} + \delta_{(\lambda_-(k), \lambda_+(k))}] \frac{dk}{2\pi}. \quad (4)$$

It is already clear that this is positive measure on one-dimensional curves.

To match the source's one-sided rectangle convention, observe that

$$\lambda'_+(k) = -\sin k \left(2s + \frac{t^2}{R(k)} \right), \quad \lambda'_-(k) = -\sin k \left(2s - \frac{t^2}{R(k)} \right). \quad (5)$$

Both derivatives are negative on $(0, \pi)$ because $R(k) \geq a$ and $2s > t^2/a$. Fix $k_0 = \pi/2$ and put $(E_1^0, E_2^0) = (\lambda_+(k_0), \lambda_-(k_0))$; this point is off the diagonal. By continuity and (5), there are constants $c, C, \eta > 0$ such that, for $0 < h < \eta$ and both signs,

$$ch \leq \lambda_{\pm}(k_0 - h) - \lambda_{\pm}(k_0) \leq Ch.$$

Hence the upper-right ε -square based at (E_1^0, E_2^0) contains the first curve in (4) for every $k = k_0 - h$ with $0 \leq h \leq \varepsilon/C$. Equation (3) is bounded below there by a positive constant, so the square has mass at least $c_0 \varepsilon$. Division by ε^2 proves the claim. $\square$

4 A completely localized Wegner example

Work on scalar $\ell^2(\mathbb{Z})$ and partition it into cells $C_n = \{2n, 2n+1\}$. Let $t > 0$, let $(\omega_n)_{n \in \mathbb{Z}}$ be i.i.d. uniform on $[-1, 1]$, and define

$$(H_\omega \psi)(2n) = \omega_n \psi(2n) + t \psi(2n+1), \quad (6)$$

$$(H_\omega \psi)(2n+1) = t \psi(2n) + \omega_n \psi(2n+1). \quad (7)$$

Thus H_ω is a bounded, self-adjoint, finite-range random Jacobi operator with deterministic hopping. Let $X \delta_j = j \delta_j$.

Let τ_a shift the cell variables and let $U_a \delta_j = \delta_{j+2a}$. Then $a \mapsto U_a$ is a unitary representation of $\mathbb{Z}$ and $U_a H_\omega U_{-a} = H_{\tau_a \omega}$ (up to the harmless choice of shift sign). The Bernoulli shift is ergodic. This is exactly a two-site fundamental cell version of the abstract covariance setup in Section 2 of [1], which assumes an arbitrary unitary representation of $\mathbb{Z}^d$. We normalize trace per physical site by

$$\mathcal{T}(A) = \frac{1}{2} \mathbb{E} \operatorname{Tr}(P_0 A P_0), \quad (8)$$

where P_0 projects onto C_0 .

Theorem 2. *For the ergodic family (6)–(7):*

1. *every spectral energy is in the complete-localization regime;*
2. *the Wegner estimate $\mathcal{T}(E_H(J)) \leq |J|/2$ holds for every Borel interval J ;*
3. *the ccc measure is*

$$M(dE_1, dE_2) = \frac{t^2}{4} \int_{-1}^1 [\delta_{(u+t, u-t)} + \delta_{(u-t, u+t)}] du; \quad (9)$$

in particular it is singular with respect to two-dimensional Lebesgue measure and has no finite density at any interior point of either off-diagonal support line.

More explicitly, for $0 < \varepsilon < \min\{1, 2t\}$,

$$\frac{M([t, t+\varepsilon], [-t, -t+\varepsilon])}{\varepsilon^2} = \frac{t^2}{4\varepsilon} \rightarrow +\infty. \quad (10)$$

Proof. On C_n , the Hamiltonian and velocity are

$$H_{\omega,n} = \begin{pmatrix} \omega_n & t \\ t & \omega_n \end{pmatrix}, \quad v_n = i[H_{\omega,n}, X] = \begin{pmatrix} 0 & it \\ -it & 0 \end{pmatrix}.$$

The velocity is independent of ω . The normalized eigenvectors

$$|+\rangle = 2^{-1/2}(1, 1), \quad |-\rangle = 2^{-1/2}(1, -1)$$

have respective energies $\omega_n + t$ and $\omega_n - t$. Direct calculation gives

$$\langle +|v_n|+ \rangle = \langle -|v_n|-\rangle = 0, \quad |\langle +|v_n|-\rangle|^2 = |\langle -|v_n|+\rangle|^2 = t^2. \quad (11)$$

Substitution of (11) into (1), followed by the cell expectation and the factor 1/2 in (8), gives (9). Its support is contained in the two lines $E_1 - E_2 = \pm 2t$, both disjoint from the diagonal.

For the rectangle in (10), the first branch of (9) contributes precisely when $u \in [0, \varepsilon]$. The transposed branch contributes nothing when $\varepsilon < 2t$. Hence its mass is $(t^2/4)\varepsilon$, proving (10). This also directly precludes a finite point density in the one-sided rectangle convention used by the source; the line support independently proves failure of absolute continuity.

It remains to verify the two regularity properties. Every eigenfunction is supported in one cell. Consequently $f(H_\omega)$ is cell-block-diagonal for every bounded Borel f , and $P_n f(H_\omega) P_m = 0$ for $n \neq m$. The strong-dynamical-localization estimate of the source therefore holds with finite constants at all energies (indeed, its left side vanishes between distinct cells).

Finally, each cell has the two eigenvalues $\omega_n \pm t$. Thus, for every interval J ,

$$\mathbb{E} \operatorname{Tr}(P_0 E_H(J) P_0) = \frac{1}{2} |([-1, 1] + t) \cap J| + \frac{1}{2} |([-1, 1] - t) \cap J| \leq |J|.$$

Dividing by two as in (8) yields $\mathcal{T}(E_H(J)) \leq |J|/2$. This proves all assertions. $\square$

5 Verification, novelty, and limitations

The accompanying standard-library script checks the block eigenpairs, velocity weights, five-cell support, four rectangle scales, the uniform-law Wegner constant, and five Floquet momenta for the signs and weight in (3)–(5). It reports **PASS**. These checks are not used as proof; all identities above are exact.

A bounded arXiv/web search on 13 August 2026 used the exact source title and the phrases “current–current correlation measure off-diagonal density,” “existence of a density current–current correlation off-diagonal,” and “current–current correlation measure random dimer.” It recovered the source and Bellissard–Hislop’s analytic-density theorem [2], but no occurrence of this cell construction or a general negative answer. The novelty assessment is therefore *apparently new within the bounded search*, not an exhaustive bibliographic guarantee.

Theorem 1 answers the question negatively at the breadth of the source’s stated scalar ergodic framework, with literal one-site translations. Theorem 2 additionally shows that complete localization plus Wegner control is insufficient in a standard unit-cell extension: a genuinely two-energy nondegeneracy assumption is needed. Neither theorem decides whether an off-diagonal density exists for the canonical Anderson Hamiltonian with independent randomness at each physical site and uniform nonzero hopping between all neighboring sites. Human review should focus on the Floquet current calculation and this scope distinction.

References

- [1] J.-M. Combes, F. Germinet, and P. D. Hislop, *Conductivity and the current–current correlation measure*, J. Phys. A: Math. Theor. **43** (2010), 474010; [arXiv:1008.5344](#).
- [2] J. V. Bellissard and P. D. Hislop, *Smoothness of correlations in the Anderson model at strong disorder*, Ann. Henri Poincaré **8** (2007), 1–26; [arXiv:math-ph/0605042](#).